SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology | Department of Agricultural Engineering

AEN 214: INTRODUCTION TO ELECTRONICS

OPEN COURSEWORK ASSIGNMENT 1: PASSIVE COMPONENTS, SEMICONDUCTOR PHYSICS & DIODE CIRCUITS

Curriculum Coverage: Topic 1 (Passive Components), Topic 2 (Semiconductor Physics), Topic 3 (Semiconductor Diodes)
Due Date / Submission: Week 5 of Semester | Individual Submission | Total Marks: 100 Marks

Instructions to Candidates:
1. This assignment is an open engineering coursework assessment designed to develop analytical and practical circuit design competence.
2. Answer ALL twelve (12) questions. Questions are tiered into Part A (Foundational Concepts), Part B (Quantitative Circuit Analysis), and Part C (Agricultural Engineering Systems Design).
3. Show all mathematical derivations, state all engineering assumptions clearly, sketch labeled schematics where applicable, and cite proper SI units.
4. Neatness, logical presentation, and clear engineering justification contribute to technical mastery and exam preparation.

PART A: Foundational Concepts & Component Physics [20 Marks]

Question 1 (5 Marks): A through-hole carbon film resistor has five colored bands in sequence: Red, Violet, Black, Brown, Brown. Decode its nominal resistance, tolerance percentage, and calculate the maximum and minimum permissible resistance values. If a 12V DC source is placed across this resistor, compute the power dissipated ($P = V^2 / R$) and state whether a 1/4W rating is adequate.

Question 2 (5 Marks): Explain the physical differences between an insulator, a semiconductor, and a conductor in terms of their energy band structures, forbidden energy gaps (E_g), and the presence of charge carriers at absolute zero (0 K) versus room temperature (300 K).

Question 3 (5 Marks): Define the term doping. Explain how adding minute traces of Phosphorus (Group V) to a pure silicon crystal produces an N-type semiconductor, and explain why the resulting crystal remains electrically neutral as a whole despite having free mobile electrons.

Question 4 (5 Marks): Distinguish between the 1st (ideal), 2nd (constant voltage drop), and 3rd (piecewise linear) approximations of a PN junction diode. State typical circuit operating conditions under which each approximation is most appropriately applied by an engineer.

PART B: Quantitative Circuit Analysis & Derivations [35 Marks]

Question 5 (8 Marks): An analog soil moisture sensor signal conditioning circuit uses a series RC low-pass filter with R = 15 kΩ and C = 4.7 μF. (a) Calculate the circuit time constant $\tau = R \cdot C$. (b) Determine the time required for the capacitor to charge from 0V to 90% of a 5V step input using $v_C(t) = V_{supply} \cdot [\ 1 - e^{-t / \tau}\ ]$. (c) Calculate the electrostatic energy stored in the capacitor $W_C = 0.5 \cdot C \cdot V^2$ when fully charged to 5V.

Question 6 (9 Marks): A monocrystalline silicon PN junction diode is fabricated with an acceptor doping concentration $N_A = 4.0 \times 10^{17}$ cm^{-3} and donor doping concentration $N_D = 1.0 \times 10^{16}$ cm^{-3} at a temperature T = 300 K (intrinsic carrier concentration $n_i = 1.5 \times 10^{10}$ cm^{-3}). (a) Calculate the junction built-in barrier potential $V_0 = V_T \cdot \ln[\ (N_A \cdot N_D) / n_i^2\ ]$. (b) If the junction temperature rises to 350 K, explain qualitatively and calculate quantitatively the directional change in V_0 (assume thermal voltage V_T scales linearly with T).

Question 7 (9 Marks): Starting from the Shockley diode equation $I_D = I_S \cdot [\ e^{V_D / (\eta V_T)} - 1\ ]$, prove mathematically that the dynamic AC junction resistance is given by $r_d = (\eta \cdot V_T) / I_D$. For a silicon diode with $\eta = 1$ and $V_T = 26$ mV, calculate r_d at: (i) $I_D = 0.5$ mA, (ii) $I_D = 2.0$ mA, and (iii) $I_D = 20$ mA.

Question 8 (9 Marks): A center-tapped transformer full-wave rectifier and a full-wave bridge rectifier are both powered from identical AC supplies. Compare both rectifiers with respect to: (a) Peak Inverse Voltage (PIV) rating required for the diodes, (b) Transformer Utilization Factor (TUF), and (c) Total forward diode voltage drop during peak conduction.

PART C: Agricultural Engineering Systems Design [45 Marks]

Question 9 (11 Marks) - Solar Irrigation Power Rectification:
Design a 12V DC power supply to operate an automated greenhouse drip-irrigation solenoid manifold from a 230 V_{rms}, 50 Hz single-phase AC generator. The four solenoid valves draw a continuous load current $I_{dc} = 2.0$ A. A full-wave bridge rectifier using four 1N5404 silicon diodes ($V_F = 0.8$ V) with a parallel reservoir capacitor filter is selected.
(a) Determine the required transformer secondary RMS voltage (V_{sec}) to maintain a minimum peak load voltage of 15.0 V.
(b) Sizing the smoothing filter capacitor C using $V_{r(pp)} = I_{dc} / (2 \cdot f \cdot C)$ to ensure peak-to-peak ripple voltage does not exceed 0.8 V_{pp}.
(c) Calculate the minimum Peak Inverse Voltage (PIV) rating required for the rectifier diodes, including a 100% safety margin ($PIV \geq 2 \cdot V_m$).
(d) Select an appropriate standard commercial capacitor value and specify its minimum working voltage (WVDC).

Question 10 (11 Marks) - Hall Effect Tractor PTO Tachometer:
A precision digital tachometer on a tractor Power Take-Off (PTO) shaft incorporates an N-type semiconductor Hall effect sensor slab with thickness t = 0.20 mm and electron carrier concentration $n = 3.2 \times 10^{15}$ cm^{-3}. A regulated bias current $I_x = 20$ mA flows longitudinally through the sensor. When a magnetic gear tooth passes within 2 mm of the sensor, a perpendicular magnetic flux density $B_z = 0.42$ Tesla is measured.
(a) Derive the expression for the Hall coefficient $R_H = 1 / (n \cdot q)$ and compute its numerical value.
(b) Calculate the generated Hall voltage $V_H = (R_H \cdot I_x \cdot B_z) / t$ in millivolts.
(c) If electrical noise from the tractor alternator induces a 2.5 mV stray voltage, calculate the Signal-to-Noise Ratio (SNR) in decibels (dB) via $SNR = 20 \cdot \log_{10}(V_{signal} / V_{noise})$ and suggest a differential amplifier topology to eliminate common-mode alternator noise.

Question 11 (11 Marks) - Sensor Analog Input Overvoltage Protection:

An analog soil conductivity probe installed along an outdoor farm furrow is connected to a 3.3V microcontroller ADC pin via a 50-meter cable. Inductive switching from nearby borehole pumps induces AC transient spikes of ±25V peak into the cable. Design a double-diode clipper protection network using two silicon diodes (V_K = 0.7 V) and a series current-limiting resistor R_S.

(a) Draw the complete circuit schematic showing connection to the 3.3V power rail and ground.

(b) Determine the maximum and minimum voltages delivered to the ADC pin during ±25V transient surges (clamped to $V_{DD} + V_K$ and $-V_K$).

(c) If the maximum allowable transient current through each protection diode is 20 mA, calculate the minimum safe value $R_S = (V_{peak} - V_{clamp}) / I_{max}$ and wattage rating of series resistor R_S.

Question 12 (12 Marks) - Capacitive Grain Moisture Sensor Filtering:

A continuous-flow maize dryer utilizes a cylindrical coaxial capacitor sensor whose dielectric constant varies linearly with grain moisture content (C_{dry} = 120 pF at 12% moisture; C_{wet} = 480 pF at 28% moisture). The sensor is excited by a 100 kHz sinusoidal oscillator.

(a) Calculate the capacitive reactance $X_C = 1 / (2 \cdot \pi \cdot f \cdot C)$ of the sensor at both dry (12%) and wet (28%) grain conditions.

(b) The sensor forms an AC voltage divider with an inductor L to convert moisture variations into amplitude variations. Determine the required inductance $L = 1 / [\ 4 \cdot \pi^2 \cdot f^2 \cdot C\ ]$ so that the circuit achieves electrical resonance at 100 kHz when grain moisture is exactly at the target storage value of 14% (C = 165 pF).

(c) What safety hazards arise if high-moisture grain causes localized dielectric heating inside the capacitor sensor?